

Practical 11 – Static Routing and RIP

AIM

To configure and verify static routing and RIP routing between networks using Cisco Packet Tracer.

REQUIREMENTS

Cisco Packet Tracer, routers, PCs, and appropriate Ethernet/serial cables.

PROCEDURE

1. Place three routers and the required PCs in the workspace.
2. Connect the devices using suitable cables.
3. Assign IP addresses to router interfaces and PCs, including default gateways.
4. Enable interfaces using the **no shutdown** command. Configure clock rate on DCE serial interfaces when required.
5. Configure static routes using **ip route** and, where required, primary and backup paths.
6. Verify static routes using **show ip route static**.
7. Enable RIP using **router rip** and advertise directly connected networks using **network** commands.
8. Test connectivity using ping and tracer.

SAMPLE STATIC ROUTE

```
ip route [destination-network] [subnet-mask] [next-hop-IP]
```

SAMPLE RIP CONFIGURATION

```
router rip
version 2
network [network-address]
no auto-summary
```

INPUT

Network addresses, subnet masks, next-hop addresses, and router interface configurations.

OUTPUT

Routing tables display the configured static routes or RIP-learned routes. Ping and tracer confirm the path and connectivity between PCs.

OBSERVATION

1. Three routers and the required PCs were interconnected in Cisco Packet Tracer, and IP addresses were assigned to all interfaces and PCs.
2. Router interfaces were enabled using **no shutdown**, and DCE serial interfaces were configured with suitable clock rate and bandwidth values.
3. Static routes with primary and backup paths were configured using **ip route** and verified using **show ip route static**.
4. RIP was enabled using **router rip**, and directly connected networks were advertised using **network**.
5. Network connectivity was verified using ping and tracer, and communication between PC0 and PC1 was successfully established.

RESULT

Static routing and RIP were configured and tested successfully in Cisco Packet Tracer.